

- Before conducting any statistical tests, exploratory data analysis will be conducted to gain an initial understanding of the data. This will facilitate future conclusions.

Overall Summary Statistics

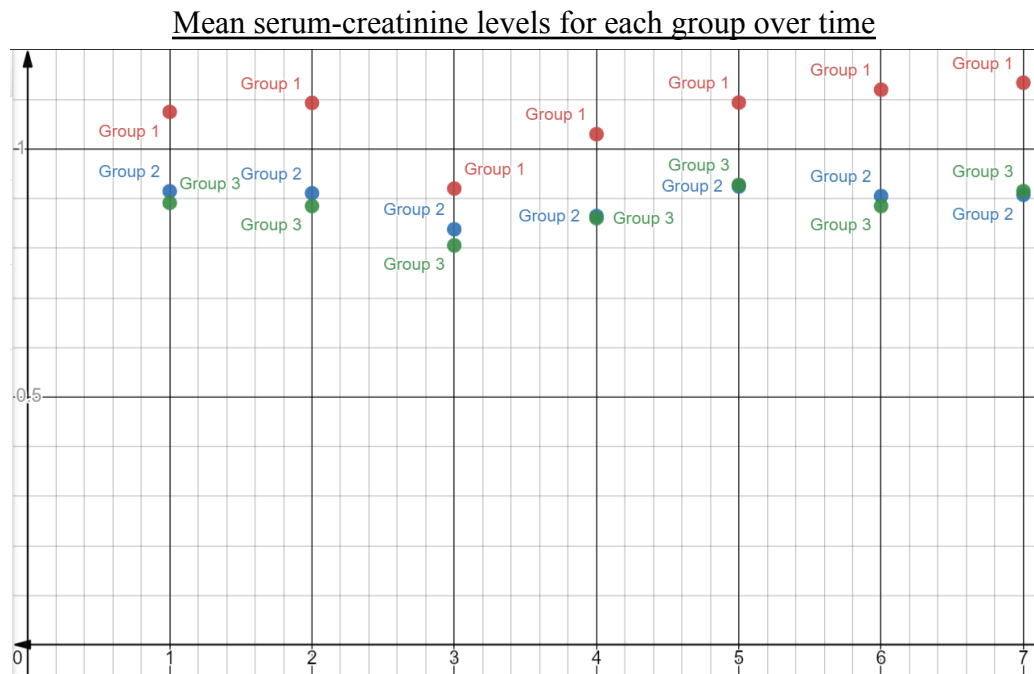
Summary of Data									
The MEANS Procedure									
Variable	N	Mean	Minimum	Lower Quartile	Median	Upper Quartile	Maximum	Range	Std Dev
age	300	41.8466667	30.0000000	37.0000000	43.0000000	46.0000000	50.0000000	20.0000000	5.5073519
baseline	295	0.9601695	0.4500000	0.7500000	0.8700000	1.0400000	3.8300000	3.3800000	0.3405792
creat_69	290	0.9631724	0.4000000	0.7200000	0.8450000	1.1000000	4.2000000	3.8000000	0.3821524
creat_70	288	0.8545139	0.4000000	0.7400000	0.8200000	0.9000000	3.7500000	3.3500000	0.2645879
creat_71	288	0.9178472	0.3700000	0.8000000	0.8800000	0.9800000	3.9000000	3.5300000	0.2729160
creat_72	290	0.9821724	0.4300000	0.8500000	0.9300000	1.0300000	5.6500000	5.2200000	0.3782611
creat_75	277	0.9709747	0.6000000	0.8200000	0.9000000	0.9900000	8.3500000	7.7500000	0.5390801
creat_78	267	0.9847566	0.6200000	0.8200000	0.9200000	1.0300000	4.8000000	4.1800000	0.3484102

It can be seen that there was loss to follow-up in this study - the total sample size decreased from 1250 to 267 by the final year. However, this should still be sufficient to draw reasonable conclusions. Additionally, it appears that the average serum-creatinine levels did not decrease over time, but this may not be important as the goal of the study was to determine any difference between groups. Thus, the overall summary statistics are not particularly informative, so the next step is to consider the summary statistics within each group. A table for these results can be found below.

Summary Statistics By Group

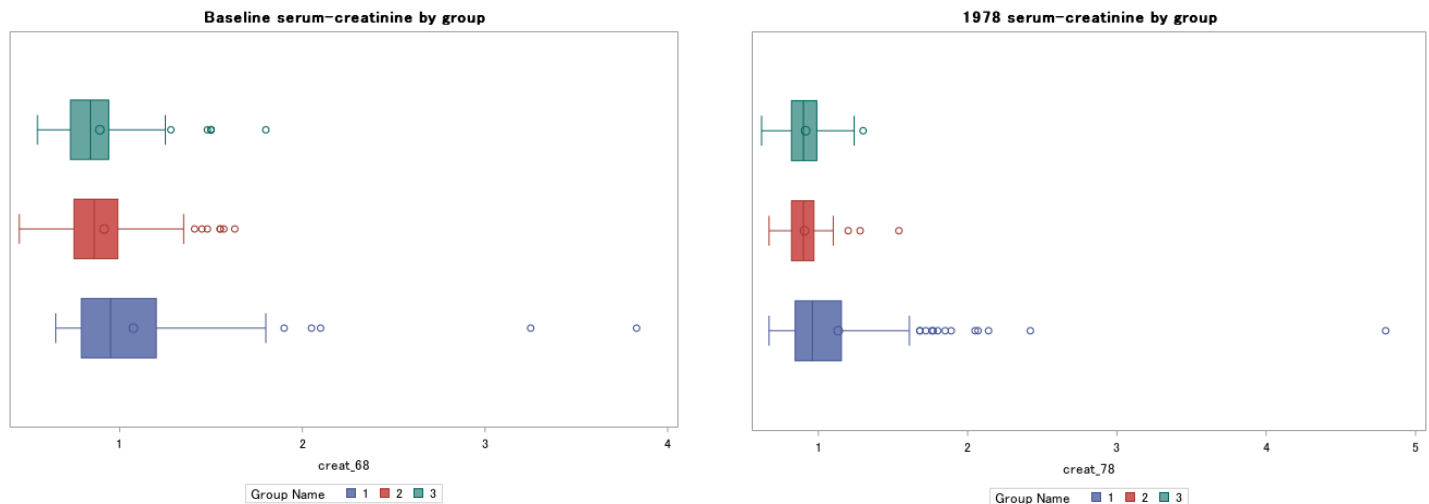
The MEANS Procedure							
group	N Obs	Variable	N	Mean	Std Dev	Minimum	Maximum
1	100	age	100	43.0000000	4.9787427	32.0000000	50.0000000
		creat_68	98	1.0747959	0.4730516	0.6500000	3.8300000
		creat_69	97	1.0929897	0.4943876	0.5600000	4.2000000
		creat_70	96	0.9203125	0.4129138	0.5700000	3.7500000
		creat_71	95	1.0298947	0.4115758	0.6900000	3.9000000
		creat_72	97	1.0937113	0.6103232	0.6400000	5.6500000
		creat_75	94	1.1195745	0.8917928	0.6700000	8.3500000
		creat_78	88	1.1337500	0.5475994	0.6700000	4.8000000
2	100	age	100	41.2300000	5.8497259	31.0000000	50.0000000
		creat_68	99	0.9147475	0.2353865	0.4500000	1.6300000
		creat_69	96	0.9109375	0.2949615	0.4000000	2.2500000
		creat_70	94	0.8381915	0.1291741	0.4000000	1.3000000
		creat_71	96	0.8647917	0.1318850	0.4500000	1.2700000
		creat_72	96	0.9245833	0.1488405	0.4300000	1.5200000
		creat_75	88	0.9050000	0.1262728	0.6100000	1.2300000
		creat_78	86	0.9076744	0.1332549	0.6700000	1.5400000
3	100	age	100	41.3100000	5.5298912	30.0000000	50.0000000
		creat_68	98	0.8914286	0.2272879	0.5500000	1.8000000
		creat_69	97	0.8850515	0.2884518	0.4200000	1.8000000
		creat_70	98	0.8057143	0.1311645	0.5200000	1.3500000
		creat_71	97	0.8606186	0.1450777	0.3700000	1.5500000
		creat_72	97	0.9276289	0.1320020	0.6700000	1.4300000
		creat_75	95	0.8850526	0.1344007	0.6000000	1.2900000
		creat_78	93	0.9150538	0.1389293	0.6200000	1.3000000

The three groups are completely balanced, albeit with some loss to follow-up within each. That is, all groups have less observations for the successive variables (recorded later in the study). The sample size for each variable between groups is rarely equal, but there does not appear to be an extreme difference in size. Additionally, it appears that the mean age in the first group (i.e. the high-NAPAP group) is slightly higher than the other two, though groups 2 and 3 have a similar mean age. Another observation is that each of the variable measurements from 1968 for 1978 decrease from group to group. That is, group 1 appears to have the highest serum-creatinine levels for each year, group 3 the lowest levels for each year, and group 2 (for the most part) between the other two levels comparatively. This can be shown in a plot for better visualization.



It is certain that group 1 has higher serum-creatinine levels than the other groups, immediately suggesting that high-NAPAP leads to higher serum-creatinine levels. Groups 2 and 3 on the other hand, low-NAPAP and the control, respectively, appear to be extremely similar. That said, the low-NAPAP group has higher serum-creatinine levels than the control for the most part, though this does not immediately appear to be significant as they surpass each other more than once. Generally speaking, there is certainly credence to the argument that high-NAPAP leads to different levels of serum-creatinine compared to the control group, but less credence to the argument that low-NAPAP does the same.

To further visualize differences between groups, box plots can be used to show the distribution of serum-creatinine levels for each group. Since creatinine levels do not appear to fluctuate much over time, the plots will look at data from 1968 (the baseline) and 1978, though analysis on data collected from the years between is certainly possible.



The baseline and final year variables appear to be similarly distributed among each group. That is, group 1 generally has higher serum-creatinine levels than the other groups, and group 3 generally has the lowest levels (although they are extremely close to the low-NAPAP for the 1978 measurements). They also appear to be dispersed somewhat evenly, though 1978 had a much larger outlier in the first group, and a much less dispersed sample in the low-NAPAP group. Both plots can support the argument that high-NAPAP is significant in causing differences in serum-creatinine, in that each of the groups are distributed closer to 0. This claim will be explored in more detail in the coming sections. For clarity, it should be reiterated that there *does not* appear to be a difference between creatinine levels for measurements taken in different years, but there *does* appear to be a difference between groups. That is, the raw data suggests that the level of NAPAP has an effect on creatinine-levels.

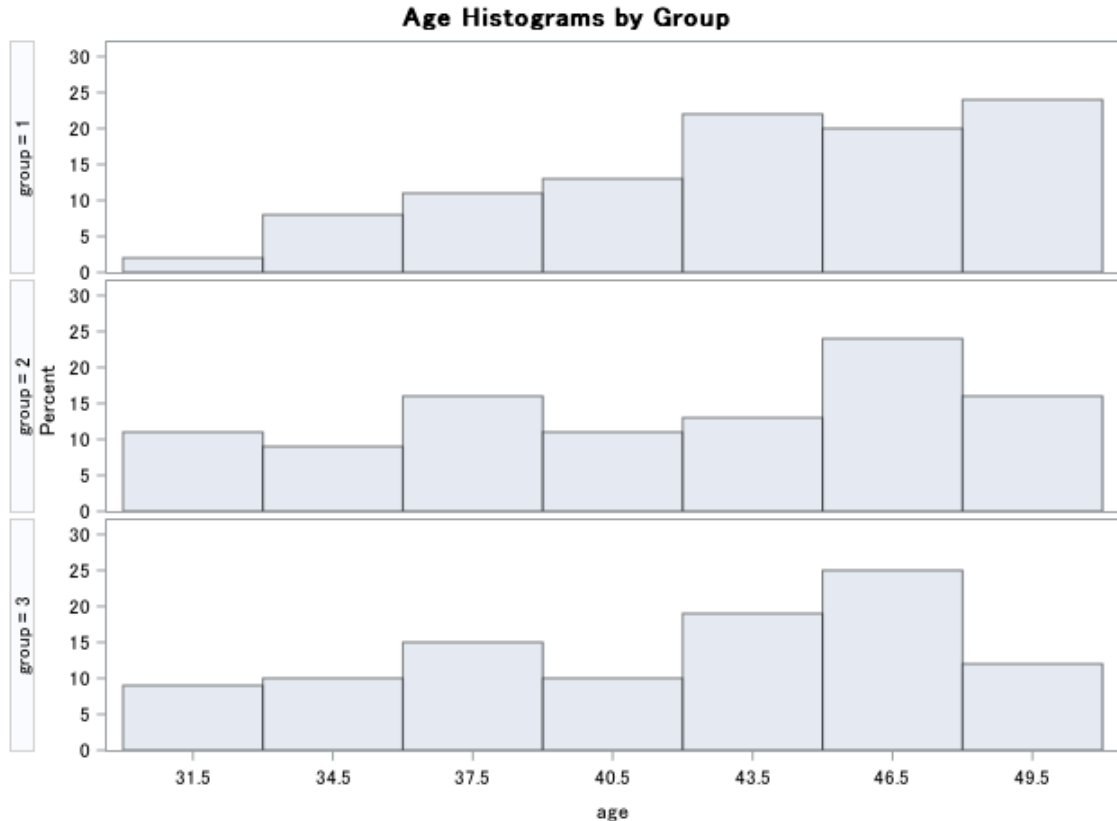
Another important factor to consider is the distribution by age for each group. Ideally, all three groups will be fairly evenly distributed to avoid the potential of age as a confound. That is, age may have an effect on the levels of serum-creatinine, so if one group largely consists of a different age range to either of the others, its results may be influenced. This could very easily lead to inaccurate conclusions regarding the effect of NAPAP on serum-creatinine levels. The SAS output for this is shown on the following page, which counts how many times each age is present within each group. This is determined without regard for the year, and instead takes the entire data set into account. Therefore, the procedure could be replicated to show the distribution of age within each group by each of the explanatory variables, which would certainly be insightful for the variables of interest (baseline and 1978).

Frequency by Age

The FREQ Procedure

Frequency Percent Row Pct Col Pct	Table of group by age																						
	group	age																					
		30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	Total
1	0	0	2	5	2	1	2	6	3	3	4	6	4	9	9	10	3	7	12	7	5	100	
	0.00	0.00	0.67	1.67	0.67	0.33	0.67	2.00	1.00	1.00	1.33	2.00	1.33	3.00	3.00	3.33	1.00	2.33	4.00	2.33	1.67	33.33	
	0.00	0.00	2.00	5.00	2.00	1.00	2.00	6.00	3.00	3.00	4.00	6.00	4.00	9.00	9.00	10.00	3.00	7.00	12.00	7.00	5.00		
	0.00	0.00	20.00	45.45	28.57	11.11	18.18	35.29	21.43	33.33	30.77	50.00	26.67	42.86	50.00	34.48	16.67	31.82	50.00	43.75	41.67		
2	0	5	6	4	1	4	6	6	4	4	4	3	5	3	5	12	6	6	6	4	6	100	
	0.00	1.67	2.00	1.33	0.33	1.33	2.00	2.00	1.33	1.33	1.33	1.00	1.67	1.00	1.67	4.00	2.00	2.00	2.00	1.33	2.00	33.33	
	0.00	5.00	6.00	4.00	1.00	4.00	6.00	6.00	4.00	4.00	4.00	3.00	5.00	3.00	5.00	12.00	6.00	6.00	6.00	4.00	6.00		
	0.00	50.00	60.00	36.36	14.29	44.44	54.55	35.29	28.57	44.44	30.77	25.00	33.33	14.29	27.78	41.38	33.33	27.27	25.00	25.00	50.00		
3	2	5	2	2	4	4	3	5	7	2	5	3	6	9	4	7	9	9	6	5	1	100	
	0.67	1.67	0.67	0.67	1.33	1.33	1.00	1.67	2.33	0.67	1.67	1.00	2.00	3.00	1.33	2.33	3.00	3.00	2.00	1.67	0.33	33.33	
	2.00	5.00	2.00	2.00	4.00	4.00	3.00	5.00	7.00	2.00	5.00	3.00	6.00	9.00	4.00	7.00	9.00	9.00	6.00	5.00	1.00		
	100.00	50.00	20.00	18.18	57.14	44.44	27.27	29.41	50.00	22.22	38.46	25.00	40.00	42.86	22.22	24.14	50.00	40.91	25.00	31.25	8.33		
Total	2	10	10	11	7	9	11	17	14	9	13	12	15	21	18	29	18	22	24	16	12	300	
	0.67	3.33	3.33	3.67	2.33	3.00	3.67	5.67	4.67	3.00	4.33	4.00	5.00	7.00	6.00	9.67	6.00	7.33	8.00	5.33	4.00	100.00	

Aside from ages 30 and 31, each group is represented by each age at least once. However, there appears to be some discrepancies between the groups. Mainly where group 1 has more persons aged above 42 than the others. Also, each group certainly has far fewer participants younger than 35 compared to age ranges 36-40, 41-45, and 46-50. In other words, the ages seem to be skewed high to some extent. A better representation of these discrepancies is the following three-row histogram.



The histogram separated the age in 3-year intervals. It can immediately be seen that the high-NAPAP group is skewed to the left, the low-NAPAP group is slightly more uniform, and the control group is less skewed left than the high-NAPAP group, but less uniform than the low-NAPAP group. Since the distribution of age appears to differ at least slightly between groups, there is a chance that the variable will have an effect on the outcome and thus the conclusions of the analysis. However, there certainly is not an extreme difference between the age distributions, and in either case there are methods which can be used to control for it as a potential covariate.

2.

ANOVA of Baseline-1968 Serum-Creatinine					
The ANOVA Procedure					
Dependent Variable: creat_68					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	2	1.95497692	0.97748846	8.88	0.0002
Error	292	32.14731461	0.11009354		
Corrected Total	294	34.10229153			

R-Square	Coeff Var	Root MSE	creat_68 Mean
0.057327	34.55676	0.331803	0.960169

Source	DF	Anova SS	Mean Square	F Value	Pr > F
group	2	1.95497692	0.97748846	8.88	0.0002

The above ANOVA table tests the hypotheses $H_0: \mu_1 = \mu_2 = \mu_3$ or $\tau_1 = \tau_2 = \tau_3 = 0$ vs. H_1 : at least one inequality.

Looking at the F-value and associated p-value, we can reject the null hypothesis and conclude that at least one baseline mean is significantly different to the others. This is not overly informative, though, because it provides no information as to *which* baseline mean differs. Therefore, an analysis on all pairwise differences can be conducted. This will be done using Tukey's procedure, as the groups are all equal in sample size. The results are on the following page.

Pairwise Differences Between Means

The ANOVA Procedure

Tukey's Studentized Range (HSD) Test for creat_68

Note: This test controls the Type I experimentwise error rate.

Alpha	0.05
Error Degrees of Freedom	292
Error Mean Square	0.110094
Critical Value of Studentized Range	3.33150

Comparisons significant at the 0.05 level
are indicated by ***.

group Comparison	Difference Between Means	Simultaneous 95% Confidence Limits		
1 – 2	0.16005	0.04867	0.27143	***
1 – 3	0.18337	0.07170	0.29503	***
2 – 1	-0.16005	-0.27143	-0.04867	***
2 – 3	0.02332	-0.08806	0.13470	
3 – 1	-0.18337	-0.29503	-0.07170	***
3 – 2	-0.02332	-0.13470	0.08806	

The output to the left supports the arguments made in the exploratory data analysis section. More specifically, it was found that there is a significant difference between the means of group 1 and group 2, and group 1 and group 3, but not a significant difference between the means of group 2 and group 3. This is to say that statistically, there is a significant difference in serum-creatinine levels between high-NAPAP and low-NAPAP *and* high-NAPAP and the control. This suggests that people with high-NAPAP are significantly more likely to have poor kidney function because their creatinine levels are higher. Alternatively, the low-NAPAP group mean and the control group mean do not differ

significantly at the 95% confidence level (moreover, their confidence intervals included a value of 0, immediately suggesting no significant difference). In the context of the study, this suggests that those with low-NAPAP levels are no more likely to have poor kidney function than those with the lowest NAPAP levels. For extra clarity, the estimated difference between means for comparisons 1-2 and 1-3 were both greater than 0, suggesting group 1 has higher serum-creatinine levels than the both of the other groups.

Baseline ANOVA					
The GLM Procedure					
Brown and Forsythe's Test for Homogeneity of creat_68 Variance ANOVA of Absolute Deviations from Group Medians					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
group	2	0.8451	0.4226	5.69	0.0038
Error	292	21.6921	0.0743		

It is important to check the assumptions of the ANOVA model before concluding the reliability of the results. The image to the left is a test for homogeneity of variance. Ideally, the null hypothesis that the three groups have the same

distribution would not be rejected. However, the SAS output shows that there is

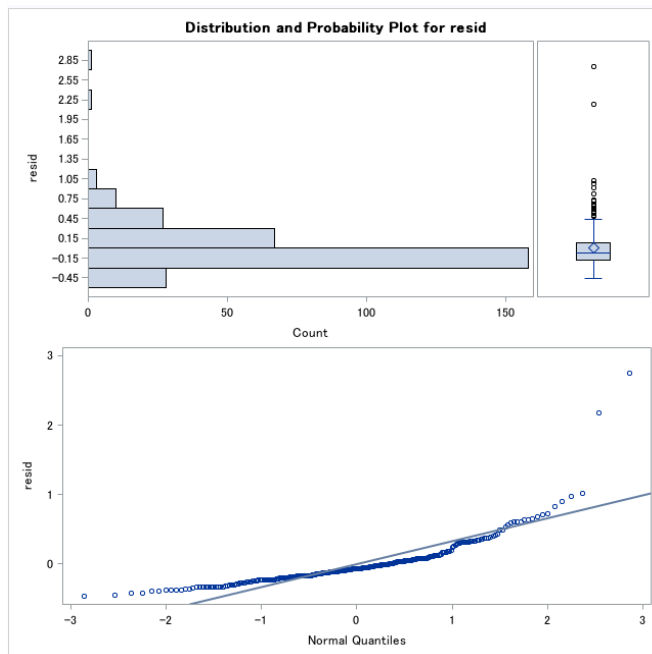
significant between-study variability, indicating the need for non-parametric ANOVA methods which will be more reliable.

The following table is the test for normality on the residuals left over by the model. Since

Tests for Normality				
Test	Statistic		p Value	
Shapiro-Wilk	W	0.746682	Pr < W	<0.0001
Kolmogorov-Smirnov	D	0.167766	Pr > D	<0.0100
Cramer-von Mises	W-Sq	2.396398	Pr > W-Sq	<0.0050
Anderson-Darling	A-Sq	13.37421	Pr > A-Sq	<0.0050

the p-values for each test statistic are very small, we reject the null hypothesis and conclude that the residuals are *not* normally distributed. This conclusion can indicate that the output variable is not normally distributed, further justifying the use of a non-parametric approach. A

histogram and plot of the residuals are included below to the left, it can be seen that they are not normally distributed simply from examining the plots.



Baseline ANOVA				
The NPAR1WAY Procedure				
Analysis of Variance for Variable creat_68 Classified by Variable group				
group	N	Mean		
1	98	1.074796		
2	99	0.914747		
3	98	0.891429		

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Among	2	1.954977	0.977488	8.8787	0.0002
Within	292	32.147315	0.110094		

Average scores were used for ties.

The above right output is from the non-parametric approach. The chi-squared test statistic is large, leading to a small p-value, which indicates that there is a significant difference between at least two of the groups as it relates to serum-creatinine levels at baseline. This conclusion agrees with the parametric approaches, so all interpretations are identical. That is, there is a significant difference between each group to group 1, but no significant difference between groups 2 and 3.

3. Repeating the procedures for serum-creatinine levels in 1978, the results are extremely similar. The ANOVA procedure yields an even lower p-value, suggesting there is at least one difference in serum-creatinine levels between the groups. This is the same conclusion as that for the baseline year, suggesting the year of measurement has little to no effect on the difference in serum-creatinine between the groups. Still, the question remains as to which groups are significantly different, so output from pairwise comparisons is included below.

1978 ANOVA
The GLM Procedure
Dependent Variable: creat_78

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	2	2.91633706	1.45816853	13.11	<.0001
Error	264	29.37332211	0.11126258		
Corrected Total	266	32.28965918			

R-Square	Coeff Var	Root MSE	creat_78 Mean
0.090318	33.87238	0.333560	0.984757

Source	DF	Type I SS	Mean Square	F Value	Pr > F
group	2	2.91633706	1.45816853	13.11	<.0001

Source	DF	Type III SS	Mean Square	F Value	Pr > F
group	2	2.91633706	1.45816853	13.11	<.0001

These results are also identical to the results from the baseline ANOVA, in that there is a significant difference in mean creatinine levels between group 1 and 2, group 1 and 3, but no significant difference between groups 2 and 3. Once again, this suggests that high-NAPAP causes significant changes to serum-creatinine levels compared to low-NAPAP and the control group. Additionally, the assumptions of ANOVA should be checked again to confirm the reliability of these results. The outputs for these checks are on the next page.

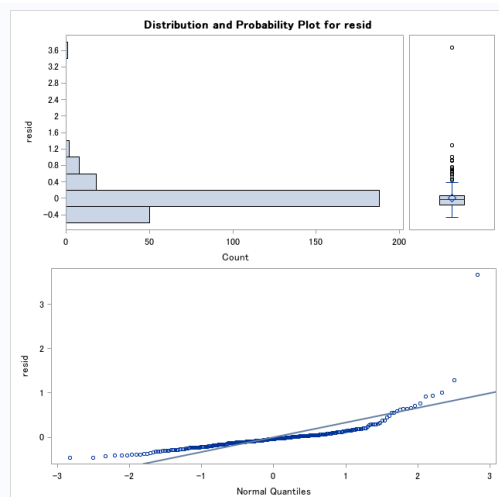
Tukey's Studentized Range (HSD) Test for creat_78

Note: This test controls the Type I experimentwise error rate.

Alpha	0.01
Error Degrees of Freedom	264
Error Mean Square	0.111263
Critical Value of Studentized Range	4.15597

Comparisons significant at the 0.01 level are indicated by *.**

group Comparison	Difference Between Means	Simultaneous 99% Confidence Limits	
1 - 3	0.21870	0.07292 0.36447	***
1 - 2	0.22608	0.07744 0.37471	***
3 - 1	-0.21870	-0.36447 -0.07292	***
3 - 2	0.00738	-0.13927 0.15402	
2 - 1	-0.22608	-0.37471 -0.07744	***
2 - 3	-0.00738	-0.15402 0.13927	



The residual plots do not appear to be normally distributed, and this is supported by the test-statistics shown in the table below. This does not satisfy one of the key assumptions of ANOVA models, so non-parametric methods should be considered. Also, the test for homogeneity resulted in another rejection of the null hypothesis, meaning the three groups are not distributed the same way, further reason to

include a non-parametric approach. The Brown-Forsythe test is shown below, in addition

Tests for Normality				
Test	Statistic		p Value	
Shapiro-Wilk	W	0.63207	Pr < W	<0.0001
Kolmogorov-Smirnov	D	0.195813	Pr > D	<0.0100
Cramer-von Mises	W-Sq	2.998312	Pr > W-Sq	<0.0050
Anderson-Darling	A-Sq	16.7913	Pr > A-Sq	<0.0050

to the non-parametric ANOVA output. As with before, the results are identical in that there is a significant difference in serum-creatinine levels between the groups.

1978 ANOVA					
The GLM Procedure					
Brown and Forsythe's Test for Homogeneity of creat_78 Variance ANOVA of Absolute Deviations from Group Medians					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
group	2	2.2473	1.1236	13.27	<.0001
Error	264	22.3621	0.0847		

1978 ANOVA					
The NPAR1WAY Procedure					
Analysis of Variance for Variable creat_78 Classified by Variable group					
group	N	Mean			
1	88	1.133750			
2	86	0.907674			
3	93	0.915054			
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Among	2	2.916337	1.458169	13.1057	<.0001
Within	264	29.373322	0.111263		
Average scores were used for ties.					

The only discernible difference between the baseline ANOVA and 1978 ANOVA is that the difference between NAPAP groups was found to be more significant in the latter, with the non-parametric F-test statistic being much larger. Thus, the p-value was smaller for the 1978 serum-creatinine levels, suggesting there is a more significant difference between group 1 and the other groups.

4. As seen from the previous problems, there is a significant difference between the high-NAPAP group and both of the other groups as it relates to serum-creatinine levels in both the baseline year and final year of the study. Additionally, the exploratory data analysis revealed that the serum-creatinine levels appeared to be somewhat constant throughout the study for each group. For that reason, it is logical to conduct an analysis on the mean change between the baseline visit and 1978 visit. The results are shown below, in addition to the pairwise differences between groups.

Mean Change ANOVA					
The GLM Procedure					
Dependent Variable: change					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	2	0.37679492	0.18839746	2.08	0.1274
Error	259	23.49813676	0.09072640		
Corrected Total	261	23.87493168			

R-Square	Coeff Var	Root MSE	change Mean
0.015782	-1448.010	0.301208	-0.020802

Source	DF	Type I SS	Mean Square	F Value	Pr > F
group	2	0.37679492	0.18839746	2.08	0.1274

Source	DF	Type III SS	Mean Square	F Value	Pr > F
group	2	0.37679492	0.18839746	2.08	0.1274

It can be seen from the output that there is no significant difference in mean change between any combination of the three groups, because the F value is relatively small, leading to a large p-value, which means the null hypothesis is not rejected. In other words, the level of NAPAP has no effect on serum-creatinine levels between the baseline visit and 1978 visit for any of the groups. Contextually, each NAPAP group had a similar change in serum-creatinine levels from the start of the study to its culmination. The table for pairwise

comparisons is included below, and shows the associated mean difference between each

group. The confidence interval for each comparison includes 0, which suggests there is no significant difference between the mean changes of any group comparison. Explicitly, the mean change of serum-creatinine levels for group 1 is not significantly different to the mean change for group 2, nor is it to group 3, nor is group 2 to group 3. As with the previous problems, it is important to check the

ANOVA assumptions to ensure the

validity of these results. SAS output and corresponding explanations follow.

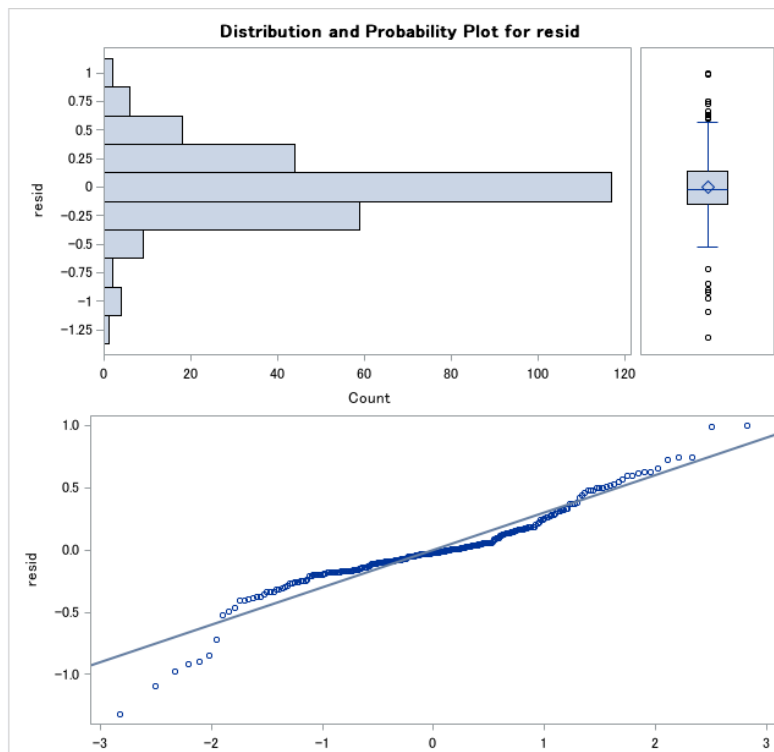
Comparisons significant at the 0.01 level are indicated by ***.			
group Comparison	Difference Between Means	Simultaneous 99% Confidence Limits	
2 - 3	0.04341	-0.09013	0.17695
2 - 1	0.09378	-0.04163	0.22919
3 - 2	-0.04341	-0.17695	0.09013
3 - 1	0.05037	-0.08277	0.18351
1 - 2	-0.09378	-0.22919	0.04163
1 - 3	-0.05037	-0.18351	0.08277

Brown and Forsythe's Test for Homogeneity of change Variance ANOVA of Absolute Deviations from Group Medians					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
group	2	0.4291	0.2145	4.36	0.0137
Error	259	12.7393	0.0492		

The first check is on the homogeneity of variance in each group. As mentioned previously, it is ideal that the three

groups have the same distribution, and thus the same variance, so that reasonable comparisons can be made. The null hypothesis is rejected in this case at $\alpha = 0.05$, meaning there is a significant difference in the variances of the three groups' mean change.

Tests for Normality				
Test	Statistic		p Value	
Shapiro-Wilk	W	0.92948	Pr < W	<0.0001
Kolmogorov-Smirnov	D	0.124712	Pr > D	<0.0100
Cramer-von Mises	W-Sq	1.009862	Pr > W-Sq	<0.0050
Anderson-Darling	A-Sq	5.712684	Pr > A-Sq	<0.0050



Additionally, an examination on the normality of residuals reveals another violation of the ANOVA assumptions, as each test statistic above is significant at $\alpha = 0.05$. Thus, it can be concluded that the residuals are *not* normally distributed, and that the model has some sort of pattern in its leftovers. This can be seen more clearly by the histogram and plot to the left. While the histogram appears to be approximately normal, the residual plot appears

almost parabolic in nature, providing a visual for the aforementioned pattern. For these reasons, a non-parametric approach should be used. It should also be mentioned that since the problem deals with a linear combination of non-normally distributed residuals, it could be expected for the combination to also be non-normal.

Mean Change ANOVA					
The NPAR1WAY Procedure					
Analysis of Variance for Variable change Classified by Variable group					
group	N	Mean			
1	86	-0.068721			
2	85	0.025059			
3	91	-0.018352			

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Among	2	0.376795	0.188397	2.0765	0.1274
Within	259	23.498137	0.090726		
Average scores were used for ties.					

The non-parametric approach results in a similar conclusion - relatively small F value leading to large p-value, so the null hypothesis that the mean change is equal among all groups is *not* rejected. This is in-line with the conclusions discussed in the parametric approach. In all, there is no significant difference between any of the three groups as it relates to mean change from the baseline visit to the 1978 visit.

- A reasonable next step would be to repeat the procedures but control for potential covariates. As mentioned in problem 1, there may be a significant difference in the mean serum-creatinine levels as a result of age. The means for group 2 and group 3 may have

Impact of NAPAP Level of Serum-Creatinine, Controlling for Age

The GLM Procedure					
Dependent Variable: creat_78					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	3.45218183	1.15072728	10.49	<.0001
Error	263	28.83747735	0.10964820		
Corrected Total	266	32.28965918			

R-Square	Coeff Var	Root MSE	creat_78 Mean
0.106913	33.62574	0.331132	0.984757

Source	DF	Type I SS	Mean Square	F Value	Pr > F
group	2	2.91633706	1.45816853	13.30	<.0001
age	1	0.53584476	0.53584476	4.89	0.0279

Source	DF	Type III SS	Mean Square	F Value	Pr > F
group	2	2.47091185	1.23545593	11.27	<.0001
age	1	0.53584476	0.53584476	4.89	0.0279

been affected by the fact their participants were on average younger than those from group 1, not to mention age was distributed differently among the three groups. This difference in mean age could be influential in serum-creatinine differences, so the process was repeated as in previous problems but controlling for age. The SAS output for the procedure is shown to the left.

It can be clearly seen that there is a significant difference between creatinine levels between at least one pair of the NAPAP groups, because the p-value in the ANOVA table

is very small. Thus the null hypothesis is rejected, and it is concluded that at least one group differs significantly to the others when controlling for age. To determine which groups differ specifically,

the least squares means can be calculated and compared.

The table shows that there is a significant difference between group 1 and group 2, and group 1 and group 3, but no significant difference between groups 2 and 3.

Once again, this is in direct agreement with the results discussed in the previous problems - high NAPAP leads to significantly different serum-creatinine levels compared to low-NAPAP and the

control. The specific means are included in the upper table, each of them different to the original. Namely, the mean for the high-NAPAP group decreased, while the low-NAPAP and control group means each increased. This essentially balanced out the means of each group, although group 1 still appears to be fairly different to the others. The alteration of the means compared to the summary statistics likely occurred because the program determined group 1 to have a significantly different distribution of age to the other groups, affecting its mean serum-creatinine value. Thus, balancing occurs after controlling for the covariate.

The output below shows the parameter estimates for the model and their respective

Parameter	Estimate		Standard Error	t Value	Pr > t
Intercept	0.5782452921	B	0.15617896	3.70	0.0003
group 1	0.2017910237	B	0.04983467	4.05	<.0001
group 2	-.0108169176	B	0.04956218	-0.22	0.8274
group 3	0.0000000000	B	.	.	.
age	0.0081826509		0.00370148	2.21	0.0279

t-values in a test for significance. The parameter estimate for the age covariate is significant at $\alpha = 0.05$, meaning it has a significant effect on serum-creatinine levels between the groups.

**The GLM Procedure
Least Squares Means**

group	creat_78 LSMEAN	Standard Error	Pr > t	LSMEAN Number
1	1.12352378	0.03560058	<.0001	1
2	0.91091583	0.03573694	<.0001	2
3	0.92173275	0.03446942	<.0001	3

**Least Squares Means for effect group
Pr > |t| for H0: LSMean(i)=LSMean(j)
Dependent Variable: creat_78**

i/j	1	2	3
1		<.0001	<.0001
2	<.0001		0.8274
3	<.0001	0.8274	

6. Next, it may be insightful to include the baseline visit as a covariate in the ANOVA model, meaning there are now two extra terms in what is now an ANCOVA model. The table below has the same conclusion as the previous ANCOVA model, with at least one group mean significantly different to the others. However, the F-value is significantly

Impact of NAPAP Level of Serum-Creatinine, Controlling for Age and Baseline Measurement

The GLM Procedure
Dependent Variable: creat_78

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	4	12.89650021	3.22412505	42.79	<.0001
Error	257	19.36541163	0.07535180		
Corrected Total	261	32.26191183			

R-Square	Coeff Var	Root MSE	creat_78 Mean
0.399744	27.87911	0.274503	0.984618

Source	DF	Type I SS	Mean Square	F Value	Pr > F
group	2	3.05054268	1.52527134	20.24	<.0001
age	1	0.58403811	0.58403811	7.75	0.0058
creat_68	1	9.26191942	9.26191942	122.92	<.0001

Source	DF	Type III SS	Mean Square	F Value	Pr > F
group	2	0.89615686	0.44807843	5.95	0.0030
age	1	0.18262853	0.18262853	2.42	0.1207
creat_68	1	9.26191942	9.26191942	122.92	<.0001

larger than any of the previous models, suggesting that the difference between group means is even more significant after controlling for baseline. This means that the values recorded in the baseline visit are extremely influential on the difference between groups. This may be trivial though, because it is

naturally expected for the baseline visit to have a significant effect on the 1978 visit because the same person is being measured both times. Thus, a high correlation between the two variables is expected, and in turn a greater explanation of the *differences* between the groups is expected because there is now an understanding of an inherent covariate in the study. The least squares means were also calculated and are included in the table below. Once again they differ from those found in the initial summary statistics for each group. Specifically, the 1978 mean for group 1 decreased, while the means for group 2 and 3 each increased, repeating the “balancing” as seen when controlling for age alone. That said, the means appear to be more balanced for this model than in the previous, another suggestion that the baseline visit is extremely impactful on the final visit. The changes to the means likely occurred because the baseline creatinine level was higher for group 1 than the other groups, in all likelihood leading to higher values in the 1978 visit. Therefore, controlling for baseline has the

The GLM Procedure
Least Squares Means

group	creat_78 LSMEAN	Standard Error	Pr > t	LSMEAN Number
1	1.06919180	0.03031072	<.0001	1
2	0.92817974	0.02985117	<.0001	2
3	0.95740909	0.02907881	<.0001	3

Least Squares Means for effect group
Pr > |t| for H0: LSMean(i)=LSMean(j)
Dependent Variable: creat_78

i/j	1	2	3
1		0.0011	0.0093
2	0.0011		0.4815
3	0.0093	0.4815	

effect of “evening out” the 1978 measurements. Also, the pairwise comparisons match which each of the procedures conducted previously, with group 1 differing significantly to both group 2 and group 3, and group 2 not differing significantly to group 3. However, the p-values for each comparison are larger. This could almost be expected though, as the

Parameter	Estimate		Standard Error	t Value	Pr > t
Intercept	0.1746305231	B	0.13553851	1.29	0.1988
group 1	0.1117827087	B	0.04264789	2.62	0.0093
group 2	-.0292293572	B	0.04146142	-0.70	0.4815
group 3	0.0000000000	B			
age	0.0048511954		0.00311610	1.56	0.1207
creat_68	0.6007852153		0.05418955	11.09	<.0001

means were all altered in such a way that made them closer together. So, a comparison between means that are close together is bound to lead to smaller test-statistics and larger p-values in turn.

One might also comment on the significance of baseline as a covariate. The table to the

left shows that the parameter associated with the covariate is extremely significant, with a very large t-value. This shows mathematically that the baseline visit is *very* influential on the difference between means among the three groups. Interestingly, the p-value for the age covariate increased a great deal, implying that the other covariate is doing the majority of the work in predicting mean differences. In other words, age is inconsequential in causing changes to serum-creatinine levels between groups relative to the baseline measurement of each group. In fact, it is more significant than all other parameters in the model.

Generally, the analysis changed in that a more significant difference between the NAPAP groups was detected. As stated, the baseline visit is *extremely* significant in affecting the 1978 measurements because the same participant is being measured both times.

Therefore its inclusion as a covariate in the model explains a very large proportion of the variability between groups.

- Contextually, it will be useful to include a variable that indicates whether a participant is more at risk for kidney malfunctioning. It is given that serum-creatinine levels over 1.3 are considered high and associated with renal insufficiency, so this variable will be controlled for in a future ANOVA.

Before conducting the analysis, it is important to look at the proportion of participants with high creatinine levels at the baseline visit within each group. A frequency table for this is included.

The FREQ Procedure

Frequency Percent Row Pct Col Pct	Table of group by high_68			
	group	high_68		
		0	1	Total
	1	83	17	100
		27.67	5.67	33.33
		83.00	17.00	
		30.74	56.67	
	2	92	8	100
		30.67	2.67	33.33
		92.00	8.00	
		34.07	26.67	
	3	95	5	100
		31.67	1.67	33.33
		95.00	5.00	
		35.19	16.67	
	Total	270	30	300
		90.00	10.00	100.00

It is clear that there are far more people with high creatinine levels in group 1 than in groups 2 or 3. This could have been anticipated for several reasons, the first being that the overall mean of group 1 was higher than the other groups, and the second being that people with greater intakes of high phenacetin-containing analgesics (as group 1 is defined) are more likely to have higher serum-creatinine levels to begin with. More specifically, 17% of women in group 1 had high serum creatinine levels, while groups 2 and 3 had 8% and 5% of their participants with high creatinine levels, respectively.

The ANCOVA table below is an analysis on the difference in mean change from baseline to 1978 between each group, controlling for those with high serum-creatinine levels. The F value was very large, leading to a rejection of the null hypothesis and conclusion that at least one mean change was significantly different to the others when controlling for high baseline creatinine levels. In the context of the problem, this means that having high creatinine levels at baseline is extremely influential on mean change over the duration of the study. This interpretation is supported by the t-value associated with the high_68 parameter. In other words, high creatinine levels at baseline explains a large proportion of the variability between mean changes among the three groups.

Impact of NAPAP Level on Serum-Creatinine, Controlling for High Baseline Measurement

The GLM Procedure

Dependent Variable: change

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	5.90350130	1.96783377	28.25	<.0001
Error	258	17.97143037	0.06965671		
Corrected Total	261	23.87493168			

R-Square	Coeff Var	Root MSE	change Mean
0.247268	-1268.780	0.263926	-0.020802

Source	DF	Type I SS	Mean Square	F Value	Pr > F
group	2	0.37679492	0.18839746	2.70	0.0688
high_68	1	5.52670638	5.52670638	79.34	<.0001

Source	DF	Type III SS	Mean Square	F Value	Pr > F
group	2	0.83134539	0.41567269	5.97	0.0029
high_68	1	5.52670638	5.52670638	79.34	<.0001

Parameter	Estimate	Standard Error	t Value	Pr > t
Intercept	-.0445249397	0.02782251	-1.60	0.1108
group 1	-.1072809736	0.04020252	-2.67	0.0081
group 2	0.0247504547	0.03986650	0.62	0.5353
group 3	0.0000000000	.	.	.
high_68	0.4763539028	0.05347835	8.91	<.0001